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Bridging UI elements across multiple operating systems

Abstract

Techniques are disclosed for enabling use of user interface (UI) elements across multiple operating systems. An indication of a UI element is selected via an interface of the first operating system. A mapping between UI elements of the first operating system and the second operating system is accessed. Based on the mapping, a corresponding UI element is provided to the second operating system. The corresponding UI element provides an equivalent UI element in the second operating system that corresponds to the UI element entered via the interface of the first operating system.

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Background/Summary

PRIORITY APPLICATION (1) This application claims the benefit of and priority to U.S. Provisional Application No. 63/398,209, filed Aug. 15, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND

(1) Virtual machines (VMs) and emulators allow a guest operating system (OS) to execute on the same computing device as a host operating system. In some cases, a feature running on the guest OS may be rendered in a proxy application running on the host OS. The proxy application may look and feel like an application that is native to the host OS. However, features native to the host operating system and features native to the guest operating system typically do not interoperate. For example, a feature for ANDROID does not automatically execute on WINDOWS. Nevertheless, there are many advantages to enabling features that are native to different operating systems to operate consistently on the same operating system. Consistently running features native to different operating systems on the same OS allows interactions between features that would otherwise be unavailable.

(2) It is with respect to these and other considerations that the disclosure made herein is presented.

SUMMARY

(3) An advantage of running features native to different operating systems (OSes) on the same OS is to increase the number of available applications and features that can run on a particular OS. However, there are some challenges in allowing features native to different OSes to run on the same OS. For example, an application running on a host OS may not be able to automate, debug, extract data from, or otherwise control a guest application that is being rendered in a proxy application running on the host OS. For example, a user may wish to control a guest application via the proxy application. However, user interface (UI) commands may not be properly forwarded from the proxy application to the guest application. Even if the UI commands are properly forwarded, the UI commands may not be properly interpreted by the guest application.

(4) One example where such incompatibilities can be problematic is in the implementation of assistant or assistive (hereinafter collectively referred to as “assistive”) technologies. Many operating systems provide an assistive service layer that provides assistive tools to enhance the way that users can interact with computing devices. Assistive tools may include computer-implemented virtual assistants, natural language and/or conversational UIs (e.g., chat UIs), artificial intelligence (AI) interfaces (e.g., those comprising large language models (LLMs) and/or other generative AI models (supporting one or more modalities), machine learning (ML), and/or other AI models). Another example of assistive tools includes computer accessibility tools that enable users with disabilities to fully interact with the computing device and provide inputs to various human-to-machine interfaces. Such computer accessibility tools can include functions such as text-to-speech and closed-captioning. Applications that allow for user inputs via assistive tools expose their UI features in order to allow the assistive service layer to access the applications and provide computer accessibility

interfaces to provide inputs and receive outputs from the applications. However, when running different operating systems, the assistive service layer that is part of one operating system and configured to interpret and render the UI features may be incompatible with applications running on a different operating system.

(5) In one example, a button on a UI may be rendered using a first operating system's assistive tools (for example to render text on the button), receive input indicating when the button has been selected (e.g., clicked/selected using an input device), and render changes to the button or transition to another UI. However, the assistive features for the same button on a second operating system may be implemented in a different manner and may not implement equivalent inputs and actions. This lack of equivalency may result in user inputs being interpreted in an inconsistent manner between the operating systems, which can lead to inconsistent actions being executed by the computing device.

(6) The techniques disclosed herein enable various UI elements, such as but not limited to assistive technologies, to be bridged between operating systems. In various embodiments, assistive tools are described that enable assistive and other applications to interact with UI controls between different operating systems. The assistive tools include components for providing automated UI functionality and can include libraries with various core functions such as screen readers. Among other benefits, the disclosed techniques enable features that do not exist in a particular operating system to be represented in that operating system, and enable those features to be changed or added in that operating system.

(7) By applying the disclosed techniques, computer processing and other resources may be used more efficiently as UI elements and other features from different operating systems may be utilized on the same device. Features and technical benefits other than those explicitly described above will be apparent from a reading of the following Detailed Description and a review of the associated drawings. This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to be used as an aid in determining the scope of the claimed subject matter. The term "techniques," for instance, may refer to system(s), method(s), computer-readable instructions, module(s), algorithms, hardware logic, and/or operation(s) as permitted by the context described above and throughout the document.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The Detailed Description is described with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The same reference numbers in different figures indicate similar or identical items. References made to individual items of a plurality of items can use a reference number with a letter of a sequence of letters to refer to each individual item. Generic references to the items may use the specific reference number without the sequence of letters.

(2) FIG. 1 is a functional diagram illustrating the disclosed embodiments.

(3) FIG. 2 illustrates an example UI element.

(4) FIG. 3 is an example functional diagram illustrating the disclosed embodiments.

(5) FIG. 4 is an example functional diagram illustrating the disclosed embodiments.

(6) FIG. 5 is an example computing environment according to an embodiment.

(7) FIG. 6 is an example computing environment according to an embodiment.

(8) FIG. 7 is an example computing environment according to an embodiment.

(9) FIG. 8 is an example operational procedure according to an embodiment.

(10) FIG. 9 is a computer architecture diagram illustrating an illustrative computer hardware and software architecture for a computing system capable of implementing aspects of the techniques and technologies presented herein.

(11) FIG. 10 is a diagram illustrating a distributed computing environment capable of implementing aspects of the techniques and technologies presented herein.

DETAILED DESCRIPTION

(12) The present disclosure describes techniques for enabling various UI elements, such as those associated with, but not limited to, assistive features, to be bridged between different operating systems. Such techniques are useful, for example, when a guest operating system is executing on the same computing device as a host operating system. Multiple operating systems may execute on the same computing device, for example, by running an emulator and a proxy application for each non-native operating system. The proxy application may look and feel like an application that is native to the host operating system. In such cases, features native to the host operating system and features native to the guest operating system may not interoperate properly. For example, a user interface (UI) element may be rendered using the host operating system's assistive tools, but a user input received via the UI element using the host operating system's assistive tools may be implemented in a different manner and may not perform equivalent inputs and/or actions in the guest operating system. In this case, the host operating system may not be able to access or otherwise process the UI element due to incompatibility of the interfaces, which can lead to inconsistent or incorrect actions being executed on the computing device. The disclosed techniques enable UI features to be bridged between different operating systems and thus operate in a consistent and correct manner.

(13) As referred to herein, a UI event refers to an operating system- (OS-) generated response to a user input, such as a mouse click event that is generated in response to a user clicking a mouse button. A UI command refers to a message sent to an application, e.g., a key press event, or a function invocation that affects an application. Generally, UI events and UI commands are referred to herein as UI elements.

(14) The interoperation of UI elements between the proxy application and the guest application is important for enabling a number of features. Merely forwarding all UI events without ensuring interoperation or otherwise modifying the UI elements may result in errors and other undesired results.

(15) To address the above and other issues, the techniques disclosed herein enable UI elements of applications running in one operating system to be bridged to another operating system instance to ensure that the UI elements are received, interpreted, and acted upon in a consistent and predictable manner. The disclosed embodiments support multiple scenarios, including assistive

features and test automation.

(16) One type of feature that can be enabled by the disclosed techniques is an assistive feature. Assistive features enhance the way users can interact with computing devices. One example of such an assistive feature includes computer accessibility features that enable users who have an impairment to interact with computer features that may otherwise be a challenge due to the user's impairment. Such assistive features may also include and be referred to as accessibility features, computer accessibility features, impairment features, or inclusivity features. Operating systems typically define a common interface for applications to expose information about the UI to enable features such as assistive features. Assistive features can include, for example, screen readers to provide information to the user about the UI and enable the user to interact with and/or manipulate the UI.

(17) Assistive features typically run in the background and receive callbacks by the system when an assistive event is detected by the operating system. Such events denote some state transition in the user interface. For example, the focus may have changed, a button may have been clicked, etc. Such an assistive feature can optionally request the capability for querying the content of the active window.

(18) The disclosed embodiments provide information about the UI elements exposed by the guest operating system to the host operating system. By exposing the information about the guest operating system's interface in a defined and consistent manner, the disclosed embodiments provide access to users of the host operating system's assistive technology in a manner that enables a seamless, native-feeling experience. Otherwise, assistive clients such as screen readers, narrators, and the like, may be blocked from providing assistive experiences when different operating systems are running in the same computing environment.

(19) FIG. 1 illustrates one example of various components that may be used to implement the disclosed UI bridging features. In an embodiment, assistive tools **110** is a set of UI automation components that are implemented in a client-side device. Assistive tools **110** can include UI automation client code **112** and UI automation core **114**. Assistive client **120** includes various application-side components that enable bridging of UI features.

(20) The components illustrated in FIG. 1 enable assistive and other features (e.g., screen readers) to interact with UI controls of various applications. Assistive tools **110** include components for providing automated UI bridging functionality and can include libraries with various core functions such as screen readers. UI automation core **114** and UI automation core **124** are OS components that include interfaces defining how various UIs are exposed and how they are accessed programmatically. UI automation core **114** and UI automation core **124** each include an application programming interface (API) **118** that enables various control frameworks to expose controls and make them visible to various client applications. API **118** enables applications running in one OS to interact with controls in other applications running in another OS and retrieve relevant information to allow for correct actions to be executed. The UI automation core **114** and UI automation core **124** host a set of libraries that enable translation of processes and provide actions that act upon a given set of UI properties.

(21) In an example, when changing a target display that is the focus of the host operating system from a virtual display to a default display, the UI automation core **114** and UI automation core **124** operate to correctly target the correct display. The default display is typically the main display that is the focus of the host operating system and renders the home screen of the guest operating system. The virtual display can represent an application in a separate window, and is typically used as an off-screen surface for applications. Without the bridging features illustrated in FIG. 1, user input actions can be sent to the default display and not the virtual display and thus applications will not have a way to send inputs to the default display. The functionality of UI automation core **114** and UI automation core **124** provide interfaces and mappings that enable applications to target the correct display.

(22) In an embodiment, assistive client **120** includes components that enable access to applications **134** running on a guest operating system **131**. The guest operating system **131** may, for example, run on a virtual machine that is hosted on a hypervisor or other virtual machine manager. The guest operating system applications **134** may each have a custom UI and expose a UI element **133** of a currently focused application to UI clients on the host operating system **100**. In an embodiment, the custom UI features of UI element **133** are mapped to control types which, as further described below, define what input controls that a given UI feature supports. In some embodiments, the custom UI features are mapped to UI mappings **126**. The UI mappings **126** can be structured with a specified layout for storage, viewing, and retrieval purposes. For example, the UI mappings **126** can be structured as a tree hierarchy.

(23) FIG. 1 illustrates a UI bridge **122** that is part of assistive client **120**. UI bridge **122** communicates with guest OS **131** via an assistive service **132** which communicates with applications **134**. For example, UI automation core **124** can query data from guest OS **131** via UI bridge **122**. If a UI being controlled by the guest OS **131** is updated by the guest OS **131**, assistive service **132** may send UI element **133** to OS assistive service **132**. UI mappings **126** provides assistive mapping information for mapping between UI elements of the host operating system **100** and the guest operating system **131**. The assistive mapping information may be used to determine a corresponding UI element **116** (referred to as UI element') to be provided to the host operating system **100**. The corresponding UI element **116** is passed to UI bridge **122** and to assistive tools **110**. The assistive mapping information in UI mappings **126**, as further described herein, may include mapping information to enable bridging of various UI features between the guest operating system **131** and those of the host operating system **100**. The assistive mapping information in UI mappings **126**, in some embodiments, can be structured as a tree.

(24) In an embodiment, a viewer tool **115** may be provided to inspect the UI elements of guest OS applications **134**. The viewer tool **115** provides a visual interface to inspect a layout hierarchy for the UI mappings **126** and view the properties of various UI components. The UI components shown in the viewer tool **115** can include those that are currently visible in the foreground of the UI of the guest OS **131**. In some cases, the guest OS **131** may provide functionality to examine details of a UI layout at runtime.

(25) In an example, use of UI elements across multiple operating systems such as host operating system **100** and a guest operating system **131** includes receiving an indication of a UI element **133** selected via an interface **136** of the guest operating system **131**, for example via a user input on a touchscreen. UI mappings **126** are accessed that include mappings between UI elements of the host operating system **100** and a guest operating system **131**. Based on the UI mappings **126**, an indication of a corresponding UI element **116** is provided to the host operating system **100**. The corresponding UI element **116** provides an equivalent UI element in the host operating system that corresponds to the UI element **133** selected via the interface **136** of the guest operating system **131**.

(26) While the described UI bridge functionality may be implemented in various components described herein, the functionality is referred generally herein as a UI automation function. In an embodiment, the UI automation function provides UI mapping

information that includes control patterns, where control types have a set of control patterns that a given UI control must support, a set that is optional, and a set that the control does not support. In an embodiment, the UI mapping information includes property values, where control types have a set of properties that the control supports.

(27) In an embodiment, the UI mapping information includes UI events, where each control type has a set of UI events that the control type supports. In an embodiment, UI mapping information is implemented as a tree structure, where each control type defines how the control appears in the UI tree structure. For UI elements that do not match a particular control type, a custom control type ID may be used to describe the control and relevant control patterns and properties.

(28) In an embodiment, the UI automation function exposes elements of a given UI to client applications as an object represented by a UI automation element interface. Each object (which may also be referred to herein as a UI element) is contained in the UI tree structure, with the desktop as the root element or a virtual root. In an embodiment, the UI automation function allows clients to view the UI tree structure or filter the UI tree structure as a control view or a content view.

(29) FIG. 2 illustrates an example of a UI automation hierarchy view. In an embodiment, the UI automation function exposes elements of the UI to client applications as an object represented by a UI automation element **220**. Elements are contained in a tree structure **200**, with the desktop **212** as the root element **210** and root element children **215** comprising application **1 216**, application **2 217**, UI automation window **218**, and so on. UI automation window **218** can further comprise automation elements text box **221** and text box **222**. Clients can filter the raw view of the tree structure **200** as a control view or a content view. The raw view includes the full tree of automation element objects for which the desktop is the root. The control view is a subset of the raw view that maps to the UI structure as viewed by a user. The content view is a subset of the control view that contains content that is relevant to the user in a given context, such as the values in a drop-down combo box. UI automation element **220** exposes properties of the control or UI element that it represents. One of these properties is the control type, which defines the basic appearance and functionality of the control or UI element as a single recognizable entity, for example, a button or check box.

(30) UI elements expose one or more control patterns. Control patterns provide a set of properties that are specific to a particular control type. Control patterns expose more information about a UI element. Mapping of control types to control patterns can be 1:1, 1 to many, or many to 1. For example, a combo box can support at least two control patterns, one to expand and collapse, and another to represent selection. Controls typically represent a single control type. UI elements are typically parts of a UI, for example a window. The UI element can define an element as a content element or a control element.

(31) In an embodiment, the UI automation function raises UI events to UI automation clients rather than broadcasting UI events. This allows for selectively raising UI events based on whether any clients are available. In an embodiment, the UI automation function includes a framework that manages child controls and rendering in an area of the screen. For example, an operating system framework can contain multiple UI automation elements, such as a menu bar, a status bar, and buttons. In an embodiment, a subtree of elements from a particular framework is referred to as a fragment. The element at the root node of the subtree is referred to as a fragment root, which can be hosted within another framework.

(32) The following provides an example implementation of the disclosed techniques using Component Object Model (COM) interfaces and WINDOWS assistive features. COM interfaces can be implemented to provide functionality for custom controls. To provide basic functionality, a UI automation provider implements at least an `IRawElementProviderSimple` interface. The `IRawElementProviderFragment` and `IRawElementProviderFragmentRoot` interfaces can be implemented for elements in a complex control to provide additional functionality. The `IRawElementProviderSimple` interface provides basic functionality for a control hosted in a window, including support for control patterns and properties. The `IRawElementProviderFragment` interface adds functionality for an element in a complex control, including navigating in the fragment, setting focus, and returning the bounding rectangle of the element. The `IRawElementProviderFragmentRoot` interface adds functionality for the root element in a complex control, including locating a child element at specified coordinates and setting the focus state for the entire control. An assistive service can be configured to receive specific types of assistive events, listen only to specific packages, receive events from each type only once in a given time frame, retrieve window content, specify a settings activity, etc.

(33) In an embodiment, the following properties are configured to receive accessibility events, which are examples of assistive events. A meta-data entry in the manifest (e.g., a document that is used for binding and activation) can be provided when declaring the service. `AccessibilityServiceInfo #eventTypes AccessibilityServiceInfo #feedbackType AccessibilityServiceInfo #flags AccessibilityServiceInfo #notificationTimeout AccessibilityServiceInfo #packageName`

(34) Window content may be retrieved which is represented as a tree of `AccessibilityWindowInfo` and `AccessibilityNodeInfo` objects using the following methods: `AccessibilityEvent.getSource( ) AccessibilityService #findFocus(int) AccessibilityService #getWindows( ) AccessibilityService #getRootInActiveWindow( )`

(35) FIG. 3 illustrates an example data flow for accessing an assistive tree. In an example, an assistive tool **320** makes a synchronous call to a component in UI automation **310**. The host OS client **312** queries the data from the guest OS client **314**. In an embodiment, the query can be a synchronous call to load the tree in a viewport. A process **313** is run to convert received messages to a tree. When the UI is updated, the host OS client **312** sends an event to the guest OS client **314**, a function **315** is run to fetch the corresponding tree and convert the tree, and the assistive tree information is sent to the host OS client **312**. The host OS client **312**

(36) FIG. 4 illustrates an example of processing assistive events including, for example, content change, button clicks, and content changes. Assistive tools **420** perform a callback to the host OS client **412** in response to a UI event. Host OS client **412** runs a process to convert messages to a node **413** and store the node. In an example, an event `TYPE_WINDOW_CONTENT_CHANGED` can be generated by assistive service **416** in UI automation function **410**. Assistive service **416** runs a process **417** to update a UI, and a UI update event is sent to guest OS client **414**. Guest OS client **414** sends an updated node **415** to host OS client **412**.

(37) The following is an example of UI automation to control mapping in an embodiment implemented between ANDROID OS and WINDOWS OS. The table illustrates an example mapping that allows for bridging between ANDROID OS and WINDOWS OS that illustrates the techniques disclosed herein, including control types and control type identifiers.

(38) TABLE-US-00001 WINDOWS Control Type ANDROID Control Type Control Type Identifier AppBar
`UIA_AppBarControlTypeId` Button `android.widget.Button` `UIA_ButtonControlTypeId` Calendar `android.widget.DatePicker`
`UIA_CalendarControlTypeId` CheckBox `android.widget.CheckBox` `UIA_CheckBoxControlTypeId` ComboBox
`android.widget.Spinner` `UIA_ComboBoxControlTypeId` DataGrid `UIA_DataGridControlTypeId` DataItem

UIA_AnimationControlTypeId Document UIA_DocumentControlTypeId Edit android.widget.EditText UIA_EditControlTypeId
 Group UIA_GroupControlTypeId Header UIA_HeaderControlTypeId HeaderItem UIA_HeaderItemControlTypeId Hyperlink
 UIA_HyperlinkControlTypeId Image android.widget.ImageView UIA_ImageControlTypeId List UIA_ListControlTypeId ListItem
 UIA_ListItemControlTypeId Menu UIA_MenuControlTypeId MenuBar UIA_MenuBarControlTypeId MenuItem
 UIA_MenuItemControlTypeId Pane UIA_PaneControlTypeId ProgressBar UIA_ProgressBarControlTypeId RadioButton
 android.widget.RadioButton UIA_RadioButtonControlTypeId ScrollBar UIA_ScrollBarControlTypeId SemanticZoom
 UIA_SemanticZoomControlTypeId Separator UIA_SeparatorControlTypeId Slider UIA_SliderControlTypeId Spinner
 UIA_SpinnerControlTypeId SplitButton UIA_SplitButtonControlTypeId StatusBar UIA_StatusBarControlTypeId Tab
 UIA_TabItemControlTypeId TabItem UIA_TabItemControlTypeId Table UIA_TableControlTypeId Text android.widget.TextView
 UIA_TextControlTypeId Thumb UIA_ThumbControlTypeId TitleBar UIA_TitleBarControlTypeId ToolBar
 UIA_ToolBarControlTypeId ToolTip UIA_ToolTipControlTypeId Tree UIA_TreeControlTypeId TreeItem
 UIA_TreeItemControlTypeId Window UIA_WindowControlTypeId

(39) While the illustrated examples make use of bridging between ANDROID and WINDOWS, the disclosed techniques may be used to bridge UI elements between any two or a plurality of operating systems. Each UI element for a first operating system may be mapped to a corresponding UI element in a second operating system. The mapping may be performed statically or dynamically. The mapping may be updated manually or updated automatically using various techniques such as machine learning.

(40) FIG. 5 illustrates an application hosting engine 550 rendering a proxy application 514 that contains content generated by a guest application 524. Computing device 502 executes host OS 510, which displays host operating desktop 552 on a display 560. Guest operating system 520 is concurrently executing on computing device 502. Guest OS 520 may be running within a hypervisor, virtual machine, emulator, or the like.

(41) Guest OS 520 renders content to guest OS desktop 522, which is also configured to render content to display 560. As such, guest OS desktop 522 and host OS desktop 552 often have the same dimensions, resolution, and other properties. Guest application 524 is running on guest OS 520. Guest application 524 renders content that would ordinarily be displayed by guest OS 520 on guest OS desktop 522. However, in order to create the impression that guest application 524 is a native application running on host OS 510 and allow increased functionality between guest application 524 and native applications running on host OS 510, application hosting engine 550 renders the content generated by guest application 524 in proxy application 514. Proxy application 514 appears on host OS desktop 552 and operates and appears as a native application of host OS 510. In some configurations, application hosting engine 550 only renders content generated by guest OS 520 that is associated with a guest application that is proxied in host OS 510. That guest OS 520 and guest application 524 do not actually render to display 560 is indicated by dashed lines of guest OS desktop 522 and guest application 524. In some configurations, guest OS 520 renders content to a pseudo display device that does not actually render content to a screen. Specifically, application hosting engine 550 provides guest application 524 with a graphics buffer into which content from guest application 524 is rendered. Proxy application 514 may then use this buffer when rendering content to host OS desktop 552.

(42) Since, as illustrated, guest OS 520 and host OS 510 are configured to use the same display 560, screen coordinates 526 and 566 are the same. As proxy application 514 is moved around host OS desktop 552, application hosting engine 550 will move guest application 524 to the corresponding location within guest OS desktop 522. Similarly, in some embodiments, as proxy application 514 is resized within host desktop 552, application hosting engine 550 will resize guest application 524 to the same width and height. In other embodiments, application hosting engine 550 may keep guest application 524 a constant size while allowing proxy application 514 to be resized. Updating the size and location of guest application 524 in this way is referred to herein as synchronizing size and location of proxy application 514 and guest application 524.

(43) In some configurations, application hosting engine 550 synchronizes the size and location of proxy application 514 and guest application 524 by forwarding UI commands received by proxy application 514, such as mouse click and mouse move events, to guest application 524. In other configurations, application hosting engine 550 maintains focus synchronicity by invoking an API to directly set the size and/or location of guest application 524.

(44) FIG. 6 illustrates an input command 646 received by the proxy application 664 rendered on host OS desktop 662 running in host operating system 610. Input command 646 may be a mouse click, keyboard input, touch input, voice command, automation command, the invocation of an operating system user interface API, or any other technique for controlling the proxy application 664. Input command 646 may be the beginning of a sequence of input commands, such as a double click, drag and drop, hover, or other user interface idiom. Guest OS 620 renders content to guest OS desktop 622. Guest application 624 is running on guest OS 620. Guest application 624 renders content on guest OS desktop 622 that would ordinarily be displayed by guest OS 620 on guest OS desktop 622.

(45) FIG. 7 illustrates the input command 746 as provided to the guest application 724 rendered on guest OS desktop 722 by the application hosting engine 750. Application hosting engine 750 provides UI command 746 to guest application 724 as input command 756. Input command 756 may be a direct copy of input command 746 or a translation to one or more corresponding UI commands recognized by applications running on guest OS 720.

(46) For example, if proxy application 774 rendered on host OS desktop 772 receives a mouse click event at a coordinate, application hosting engine 750 may create a mouse click event with the coordinate but that is formatted for applications running on guest OS 720. In some configurations, application hosting engine 750 may then forward the newly created event to guest application 724. In other configurations, application hosting engine 750 forwards the newly created event to guest OS 720, which forwards it to guest application 724. Guest OS 720 may forward the newly created event to guest application 724 because guest application 724 is the focus of guest OS 720, or because the newly created event specifically targets guest application 724. Instead of forwarding the mouse click event, application hosting engine 750 may invoke an automation API, OS API, or an application-specific API to cause guest application 724 to perform a function equivalent to the function that mouse click was expected to perform.

(47) In one example, a first or primary OS is the host OS 710 of a hypervisor 770 and a secondary OS is the guest OS 720 running on the hypervisor 770. The hypervisor 770 is typically a software-based manager for creating and running virtual machines. Instead of displaying the entire desktop of the guest OS 720, an individual application running on the guest OS 720 may be virtualized, appearing on the desktop of the primary OS. Specifically, a proxy application 774 running on the primary OS renders the user

interface of a guest application **724** running on the secondary OS. User interface events received by the proxy application **774** are forwarded to the guest application **724**. In this way, the guest application **724** appears and operates as a native application within the desktop of the primary OS. While the example of a hypervisor concurrently running two operating systems on the same device is used throughout this disclosure, any other configurations in which UI elements are bridged between operating systems are also contemplated.

(48) In some configurations, an application hosting engine **750** instantiates the proxy application **774**, configures the proxy application **774** to render the UI content of the guest application **724**, and provides UI commands received by the proxy application **774** to the guest application **724**. The application hosting engine **750** may translate or reformat UI commands received by the proxy application **774** to the equivalent UI commands of the secondary OS. For example, when the proxy application **774** receives a primary OS mouse click event, the application hosting engine **750** will translate or reformat the primary OS mouse click event as necessary to create a secondary OS mouse click event. The application hosting engine then forwards the secondary OS mouse click event to the secondary OS. The secondary OS forwards the secondary OS mouse click event to the application with the focus of the secondary OS.

(49) Generally, an operating system provides focus to one application at a time, referring to the operating system only supporting direct user interaction with one application at a time. Specifically, user input from a keyboard, mouse, finger, voice or other input mechanism is sent to a focused application, which can also be referred to as a foreground application or an active application. The focused application is typically distinguished visually from background applications that do not receive user input. Setting the focus of an operating system refers to selecting which application receives user input. As used herein, references to a focused application also refer to a focused window, or an aspect of a GUI the operating system directs user input to.

(50) Turning now to FIG. **8**, illustrated are aspects of a routine for enabling use of user UI elements across multiple operating systems. For ease of understanding, the processes discussed in this disclosure are delineated as separate operations represented as independent blocks. However, these separately delineated operations should not be construed as necessarily order dependent in their performance. The order in which the process is described is not intended to be construed as a limitation, and any number of the described process blocks may be combined in any order to implement the process or an alternate process. Moreover, it is also possible that one or more of the provided operations is modified or omitted.

(51) With reference to FIG. **8**, routine **800** begins at operation **802**, which illustrates running at least a first operating system and a second operating system in a virtualized computing system.

(52) Operation **804** illustrates receiving an indication of a UI element selected via an interface at the first operating system.

(53) Operation **806** illustrates accessing a mapping between UI elements of the first operating system and the second operating system.

(54) Operation **808** illustrates based on the mapping, providing a corresponding UI element to the second operating system. The corresponding UI element provides an equivalent UI element in the second operating system that corresponds to the UI element entered via the interface at the first operating system.

(55) The particular implementation of the technologies disclosed herein is a matter of choice dependent on the performance and other requirements of a computing device. Accordingly, the logical operations described herein are referred to variously as states, operations, structural devices, acts, or modules. These states, operations, structural devices, acts, and modules can be implemented in hardware, software, firmware, in special-purpose digital logic, and any combination thereof. It should be appreciated that more or fewer operations can be performed than shown in the figures and described herein. These operations can also be performed in a different order than those described herein.

(56) It also should be understood that the illustrated methods can end at any time and need not be performed in their entirety. Some or all operations of the methods, and/or substantially equivalent operations, can be performed by execution of computer-readable instructions included on a computer-storage media, as defined below. The term “computer-readable instructions,” and variants thereof, as used in the description and claims, is used expansively herein to include routines, applications, application modules, program modules, programs, components, data structures, algorithms, and the like. Computer-readable instructions can be implemented on various system configurations, including single-processor or multiprocessor systems, minicomputers, mainframe computers, personal computers, hand-held computing devices, microprocessor-based, programmable consumer electronics, combinations thereof, and the like.

(57) Thus, it should be appreciated that the logical operations described herein are implemented (1) as a sequence of computer implemented acts or program modules running on a computing system and/or (2) as interconnected machine logic circuits or circuit modules within the computing system. The implementation is a matter of choice dependent on the performance and other requirements of the computing system. Accordingly, the logical operations described herein are referred to variously as states, operations, structural devices, acts, or modules. These operations, structural devices, acts, and modules may be implemented in software, in firmware, in special purpose digital logic, and any combination thereof.

(58) For example, the operations of the routine **800** are described herein as being implemented, at least in part, by modules running the features disclosed herein can be a dynamically linked library (DLL), a statically linked library, functionality produced by an application programming interface (API), a compiled program, an interpreted program, a script or any other executable set of instructions. Data can be stored in a data structure in one or more memory components. Data can be retrieved from the data structure by addressing links or references to the data structure.

(59) Although the following illustration refers to the components of the figures, it should be appreciated that the operations of the routine **800** may be also implemented in many other ways. For example, the routine **800** may be implemented, at least in part, by a processor of another remote computer or a local circuit. In addition, one or more of the operations of the routine **800** may alternatively or additionally be implemented, at least in part, by a chipset working alone or in conjunction with other software modules. In the example described below, one or more modules of a computing system can receive and/or process the data disclosed herein. Any service, circuit or application suitable for providing the techniques disclosed herein can be used in operations described herein.

(60) FIG. **9** shows additional details of an example computer architecture **900** for a device, such as a computer or a server configured as part of the systems described herein, capable of executing computer instructions (e.g., a module or a program component

described herein) the computer architecture **900** illustrated in FIG. 9 includes processing unit(s) **902**, a system memory **904**, including a random-access memory **906** (“RAM”) and a read-only memory (“ROM”) **908**, and a system bus **910** that couples the memory **904** to the processing unit(s) **902**.

(61) Processing unit(s), such as processing unit(s) **902**, can represent, for example, a CPU-type processing unit, a GPU-type processing unit, a field-programmable gate array (FPGA), another class of digital signal processor (DSP), or other hardware logic components that may, in some instances, be driven by a CPU. For example, and without limitation, illustrative types of hardware logic components that can be used include Application-Specific Integrated Circuits (ASICs), Application-Specific Standard Products (ASSPs), System-on-a-Chip Systems (SOCs), Complex Programmable Logic Devices (CPLDs), etc.

(62) A basic input/output system containing the basic routines that help to transfer information between elements within the computer architecture **900**, such as during startup, is stored in the ROM **908**. The computer architecture **900** further includes a mass storage device **912** for storing an operating system **914**, application(s) **916**, modules **918**, and other data described herein.

(63) The mass storage device **912** is connected to processing unit(s) **902** through a mass storage controller connected to the bus **910**. The mass storage device **912** and its associated computer-readable media provide non-volatile storage for the computer architecture **900**. Although the description of computer-readable media contained herein refers to a mass storage device, it should be appreciated by those skilled in the art that computer-readable media can be any available computer-readable storage media or communication media that can be accessed by the computer architecture **900**.

(64) Computer-readable media can include computer-readable storage media and/or communication media. Computer-readable storage media can include one or more of volatile memory, nonvolatile memory, and/or other persistent and/or auxiliary computer storage media, removable and non-removable computer storage media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data. Thus, computer storage media includes tangible and/or physical forms of media included in a device and/or hardware component that is part of a device or external to a device, including but not limited to random access memory (RAM), static random-access memory (SRAM), dynamic random-access memory (DRAM), phase change memory (PCM), read-only memory (ROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, compact disc read-only memory (CD-ROM), digital versatile disks (DVDs), optical cards or other optical storage media, magnetic cassettes, magnetic tape, magnetic disk storage, magnetic cards or other magnetic storage devices or media, solid-state memory devices, storage arrays, network attached storage, storage area networks, hosted computer storage or any other storage memory, storage device, and/or storage medium that can be used to store and maintain information for access by a computing device.

(65) In contrast to computer-readable storage media, communication media can embody computer-readable instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave, or other transmission mechanism. As defined herein, computer storage media does not include communication media. That is, computer-readable storage media does not include communications media consisting solely of a modulated data signal, a carrier wave, or a propagated signal, per se.

(66) According to various configurations, the computer architecture **900** may operate in a networked environment using logical connections to remote computers through the network **920**. The computer architecture **900** may connect to the network **920** through a network interface unit **922** connected to the bus **910**. The computer architecture **900** also may include an input/output controller **924** for receiving and processing input from a number of other devices, including a keyboard, mouse, touch, or electronic stylus or pen. Similarly, the input/output controller **924** may provide output to a display screen, a printer, or other type of output device.

(67) It should be appreciated that the software components described herein may, when loaded into the processing unit(s) **902** and executed, transform the processing unit(s) **902** and the overall computer architecture **900** from a general-purpose computing system into a special-purpose computing system customized to facilitate the functionality presented herein. The processing unit(s) **902** may be constructed from any number of transistors or other discrete circuit elements, which may individually or collectively assume any number of states. More specifically, the processing unit(s) **902** may operate as a finite-state machine, in response to executable instructions contained within the software modules disclosed herein. These computer-executable instructions may transform the processing unit(s) **902** by specifying how the processing unit(s) **902** transition between states, thereby transforming the transistors or other discrete hardware elements constituting the processing unit(s) **902**.

(68) FIG. 10 depicts an illustrative distributed computing environment **1000** capable of executing the software components described herein. Thus, the distributed computing environment **1000** illustrated in FIG. 10 can be utilized to execute any aspects of the software components presented herein. For example, the distributed computing environment **1000** can be utilized to execute aspects of the software components described herein.

(69) Accordingly, the distributed computing environment **1000** can include a computing environment **1002** operating on, in communication with, or as part of the network **1004**. The network **1004** can include various access networks. One or more client devices **1006A-1006N** (hereinafter referred to collectively and/or generically as “clients **1006**” and also referred to herein as computing devices **1006**) can communicate with the computing environment **1002** via the network **1004**. In one illustrated configuration, the clients **1006** include a computing device **1006A** such as a laptop computer, a desktop computer, or other computing device; a slate or tablet computing device (“tablet computing device”) **1006B**; a mobile computing device **1006C** such as a mobile telephone, a smart phone, or other mobile computing device; a server computer **1006D**; and/or other devices **1006N**. It should be understood that any number of clients **1006** can communicate with the computing environment **1002**.

(70) In various examples, the computing environment **1002** includes servers **1008**, data storage **1010**, and one or more network interfaces **1012**. The servers **1008** can host various services, virtual machines, portals, and/or other resources. In the illustrated configuration, the servers **1008** host virtual machines **1014**, Web portals **1016**, mailbox services **1018**, storage services **1020**, and/or, social networking services **1022**. As shown in FIG. 6 the servers **1008** also can host other services, applications, portals, and/or other resources (“other resources”) **1024**.

(71) As mentioned above, the computing environment **1002** can include the data storage **1010**. According to various implementations, the functionality of the data storage **1010** is provided by one or more databases operating on, or in communication with, the network **1004**. The functionality of the data storage **1010** also can be provided by one or more servers configured to host data for the computing environment **1002**. The data storage **1010** can include, host, or provide one or more real or virtual datastores **1026A-1026N** (hereinafter referred to collectively and/or generically as “datastores **1026**”). The datastores **1026** are configured to

host data used or created by the servers **1008** and/or other data. That is, the datastores **1026** also can host or store web page documents, word documents, presentation documents, data structures, algorithms for execution by a recommendation engine, and/or other data utilized by any application program. Aspects of the datastores **1026** may be associated with a service for storing files.

(72) The computing environment **1002** can communicate with, or be accessed by, the network interfaces **1012**. The network interfaces **1012** can include various types of network hardware and software for supporting communications between two or more computing devices including, but not limited to, the computing devices and the servers. It should be appreciated that the network interfaces **1012** also may be utilized to connect to other types of networks and/or computer systems.

(73) It should be understood that the distributed computing environment **1000** described herein can provide any aspects of the software elements described herein with any number of virtual computing resources and/or other distributed computing functionality that can be configured to execute any aspects of the software components disclosed herein. According to various implementations of the concepts and technologies disclosed herein, the distributed computing environment **1000** provides the software functionality described herein as a service to the computing devices. It should be understood that the computing devices can include real or virtual machines including, but not limited to, server computers, web servers, personal computers, mobile computing devices, smart phones, and/or other devices. As such, various configurations of the concepts and technologies disclosed herein enable any device configured to access the distributed computing environment **1000** to utilize the functionality described herein for providing the techniques disclosed herein, among other aspects.

(74) While certain example embodiments have been described, these embodiments have been presented by way of example only and are not intended to limit the scope of the inventions disclosed herein. Thus, nothing in the foregoing description is intended to imply that any particular feature, characteristic, step, module, or block is necessary or indispensable. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions disclosed herein. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of certain of the inventions disclosed herein.

(75) It should be appreciated that any reference to “first,” “second,” etc. elements within the Summary and/or Detailed Description is not intended to and should not be construed to necessarily correspond to any reference of “first,” “second,” etc. elements of the claims. Rather, any use of “first” and “second” within the Summary, Detailed Description, and/or claims may be used to distinguish between two different instances of the same element.

(76) In closing, although the various techniques have been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended representations is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as example forms of implementing the claimed subject matter.

(77) The disclosure presented herein also encompasses the subject matter set forth in the following clauses:

(78) Clause 1: A method for enabling use of user interface (UI) elements across multiple operating systems, the method comprising: running at least a first operating system and a second operating system in a virtualized computing system; receiving an indication of a UI element selected via an interface of the first operating system; accessing a mapping between UI elements of the first operating system and the second operating system; and based on the mapping, providing a corresponding UI element to the second operating system.

(79) Clause 2: The method of clause 1, wherein receiving the indication of the UI element comprises receiving the indication of the UI element via a user interface generated by a guest application and rendered in a desktop of the first operating system.

(80) Clause 3: The method of any of clauses 1-2, wherein receiving the indication of the UI element comprises receiving a control type and a control pattern associated with the UI element.

(81) Clause 4: The method of any of clauses 1-3, wherein receiving the control type and the control pattern associated with the UI element comprises receiving a set of properties that are specific to a particular control type.

(82) Clause 5: The method of any of clauses 1-4, wherein receiving the indication of the UI element comprises receiving a tree structure that includes the indication of the UI element.

(83) Clause 6: The method of any of clauses 1-5, wherein the UI element is for an assistive feature.

(84) Clause 7: The method of clauses 1-6, wherein running the first operating system comprises running the first operating system on an emulator operating in the virtualized computing system.

(85) Clause 8: The method of any of clauses 1-7, further comprising instantiating an application programming interface (API) configured to expose custom controls to applications and receive requests for information about the applications.

(86) Clause 9: The method of any of clauses 1-8, further comprising instantiating a UI bridge configured to communicate with the first operating system via an assistive service of the first operating system that communicates with applications running on the first operating system.

(87) Clause 10: A system, comprising: one or more processors; and a computer-readable storage medium having encoded thereon computer-executable instructions to cause the one or more processors to perform operations comprising: receiving an indication of a UI element selected via an interface of a first operating system, the UI element comprising a control type and control pattern; accessing a mapping between UI elements of the first operating system and a second operating system, the UI elements of the mapping comprising a control type and control pattern; and based on the mapping, providing a corresponding UI element to the second operating system.

(88) Clause 11: The system of clause 10, wherein the UI element is contained in a tree structure.

(89) Clause 12: The system of any of clauses 10 and 11, wherein the UI element is for an assistive feature.

(90) Clause 13: The system of any of clauses 10-12, wherein the first operating system is configured to run in an emulator operating in the system.

(91) Clause 14: The system of any of clauses 10-13, further comprising computer-executable instructions that, when executed by the one or more processors, cause the system to perform operations comprising instantiating an application programming interface (API) configured to expose custom controls to applications and receive requests for information about the applications.

(92) Clause 15: The system of any of clauses 10-14, further comprising computer-executable instructions that, when executed by the

one or more processors, cause the system to perform operations comprising instantiating a UI bridge configured to communicate with the first operating system via an assistive service of the first operating system that communicates with applications running on the first operating system.

(93) Clause 16: A computer-readable storage medium having computer-executable instructions stored thereupon which, when executed by one or more processors of a computing device, cause the computing device to perform operations comprising: receiving a UI element selected via an interface of a first operating system, the UI element comprising a control type and control pattern; accessing a mapping between UI elements of the first operating system and a second operating system, the UI elements of the mapping comprising a control type and control pattern; and based on the mapping, providing a corresponding UI element to the second operating system.

(94) Clause 17: The computer-readable storage medium of clause 16, further comprising computer-executable instructions that, when executed by the one or more processors, cause the computing device to perform operations comprising instantiating an application programming interface (API) configured to expose custom controls to applications and receive requests for information about the applications.

(95) Clause 18: The computer-readable storage medium of any of clauses 16 and 17, wherein the UI element comprises a control type and control pattern.

(96) Clause 19: The computer-readable storage medium of any of clauses 16-18, wherein the UI element is contained in a tree structure.

(97) Clause 20: The computer-readable storage medium of any of clauses 16-19, further comprising computer-executable instructions that, when executed by the one or more processors, cause the computing device to perform operations comprising instantiating a user interface (UI) bridge configured to communicate with the first operating system via an assistive service of the first operating system that communicates with applications running on the first operating system.

Claims

1. A method for enabling use of user interface (UI) elements across multiple operating systems, the method comprising: running at least a first operating system and a second operating system in a virtualized computing system; receiving an indication of a UI element selected via an interface of the first operating system; accessing a mapping between UI elements of the first operating system and the second operating system; wherein: custom UI features of the UI elements of the first operating system are mapped to control types defining input controls that a given UI feature supports, the control types indicative of appearance and functionality of the UI elements as a single entity; wherein the mapping includes: control patterns where the control patterns define functional behavior of a UI element and where the control types are indicative of: control patterns that a given UI control supports, control patterns that are optional, or control patterns that are not supported; property values for the control types, the property values indicative of UI properties that the given UI control supports; and UI events indicative of operating system-generated responses to user inputs, wherein each control type comprises a set of UI events that the control type supports; and based on the mapping, providing a corresponding UI element to the second operating system.

2. The method of claim 1, wherein receiving the indication of the UI element comprises receiving the indication of the UI element via a user interface generated by a guest application and rendered in a desktop of the first operating system.

3. The method of claim 1, wherein receiving the indication of the UI element comprises receiving a tree structure that includes the indication of the UI element.

4. The method of claim 1, wherein the UI element is for an assistive feature.

5. The method of claim 1, wherein running the first operating system comprises running the first operating system on an emulator operating in the virtualized computing system.

6. The method of claim 1, further comprising instantiating an application programming interface (API) configured to expose custom controls to applications and receive requests for information about the applications.

7. The method of claim 1, further comprising instantiating a UI bridge configured to communicate with the first operating system via an assistive service of the first operating system that communicates with applications running on the first operating system.

8. A system, comprising: one or more processors; and a computer-readable storage medium having encoded thereon computer-executable instructions to cause the one or more processors to perform operations comprising: receiving an indication of a UI element selected via an interface of a first operating system; accessing a mapping between UI elements of the first operating system and a second operating system; wherein: custom UI features of the UI elements of the first operating system are mapped to control types defining input controls that a given UI feature supports, the control types indicative of appearance and functionality of the UI elements as a single entity; wherein the mapping includes: control patterns where the control patterns define functional behavior of a UI element and where the control types are indicative of: control patterns that a given UI control supports, control patterns that are optional, or control patterns that are not supported; property values for the control types, the property values indicative of UI properties that the given UI control supports; and UI events indicative of operating system-generated responses to user inputs, wherein each control type comprises a set of UI events that the control type supports; and based on the mapping, providing a corresponding UI element to the second operating system.

9. The system of claim 8, wherein the UI element is contained in a tree structure.

10. The system of claim 8, wherein the UI element is for an assistive feature.

11. The system of claim 8, wherein the first operating system is configured to run in an emulator operating in the system.

12. The system of claim 8, further comprising computer-executable instructions that, when executed by the one or more processors, cause the system to perform operations comprising instantiating an application programming interface (API) configured to expose custom controls to applications and receive requests for information about the applications.

13. The system of claim 8, further comprising computer-executable instructions that, when executed by the one or more processors, cause the system to perform operations comprising instantiating a UI bridge configured to communicate with the first operating system via an assistive service of the first operating system that communicates with applications running on the first operating

system.

14. A computer-readable storage medium having computer-executable instructions stored thereupon which, when executed by one or more processors of a computing device, cause the computing device to perform operations comprising: receiving a UI element selected via an interface of a first operating system; accessing a mapping between UI elements of the first operating system and a second operating system; wherein: custom UI features of the UI elements of the first operating system are mapped to control types defining input controls that a given UI feature supports, the control types indicative of appearance and functionality of the UI elements as a single entity; wherein the mapping includes: control patterns where the control patterns define functional behavior of a UI element and where the control types are indicative of: control patterns that a given UI control supports, control patterns that are optional, or control patterns that are not supported; property values for the control types, the property values indicative of UI properties that the given UI control supports; and UI events indicative of operating system-generated responses to user inputs, wherein each control type comprises a set of UI events that the control type supports; and based on the mapping, providing a corresponding UI element to the second operating system.

15. The computer-readable storage medium of claim 14, further comprising computer-executable instructions that, when executed by the one or more processors, cause the computing device to perform operations comprising instantiating an application programming interface (API) configured to expose custom controls to applications and receive requests for information about the applications.

16. The computer-readable storage medium of claim 14, wherein the UI element is contained in a tree structure.

17. The computer-readable storage medium of claim 16, further comprising computer-executable instructions that, when executed by the one or more processors, cause the computing device to perform operations comprising instantiating a user interface (UI) bridge configured to communicate with the first operating system via an assistive service of the first operating system that communicates with applications running on the first operating system.
